



Skills proven, not claimed.

SkillProof turns a student profile into a measurable readiness score and an adaptive roadmap, with the evidence behind every number.



The problem

A student knows the role they want. Nobody tells them what they are missing for it.

Resumes are self reported

a listed skill and a proven skill look identical on paper

Roadmaps are generic

the same eight week plan for a student with four shipped projects and one with none

Nothing is measurable

no number to argue with, so no way to know if this month helped

The result

months on the wrong skill, and a portfolio that does not answer the recruiter question

Why existing tools fall short

Tool	What it does	What it misses
Course platforms	Sell a fixed syllabus	No idea what you already know
Resume scanners	Match keywords to a job description	Trusts the resume completely
A chatbot	Writes a plausible roadmap	No evidence, no score, no memory, invented links
Placement portals	List openings	Never say what you are missing for them



SkillProof

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Skills proven, not claimed. Four steps, all visible to the student.

Collect evidence

resume gives claimed skills, GitHub code gives observed skills traced to the file, a quiz gives verified skills

Price the gaps

readiness is the job description weighted share of the role you can prove

Plan against the evidence

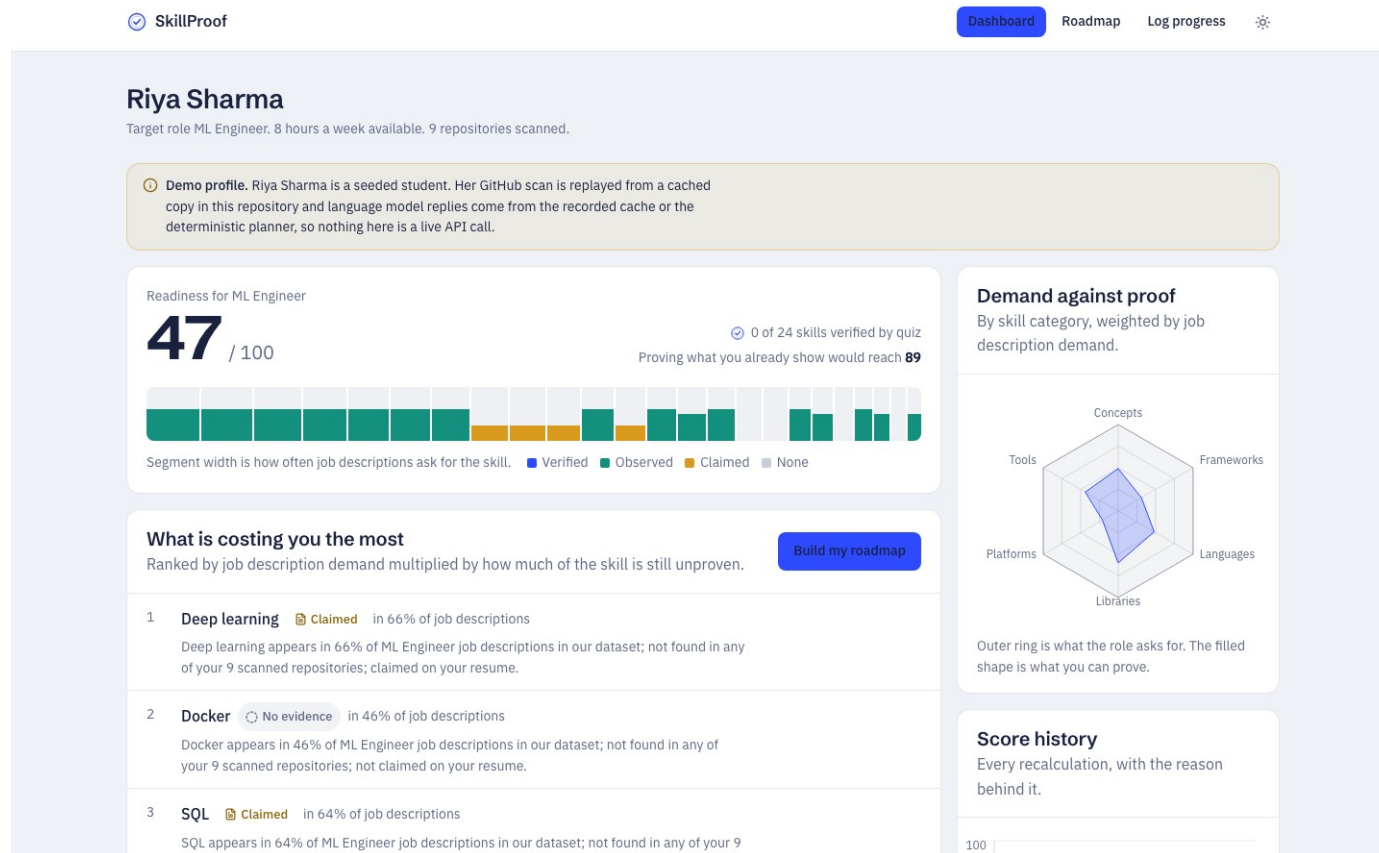
ranked by cost, ordered by prerequisite, packed into your hours, ending in a proof project

Replan on proof

every new piece of evidence recalculates the score and rebuilds the weeks you have not touched

One number, and the evidence behind it

Readiness 47 out of 100 for ML Engineer. Every segment is a skill the role asks for.



Segment width is job description demand

Colour is the evidence level

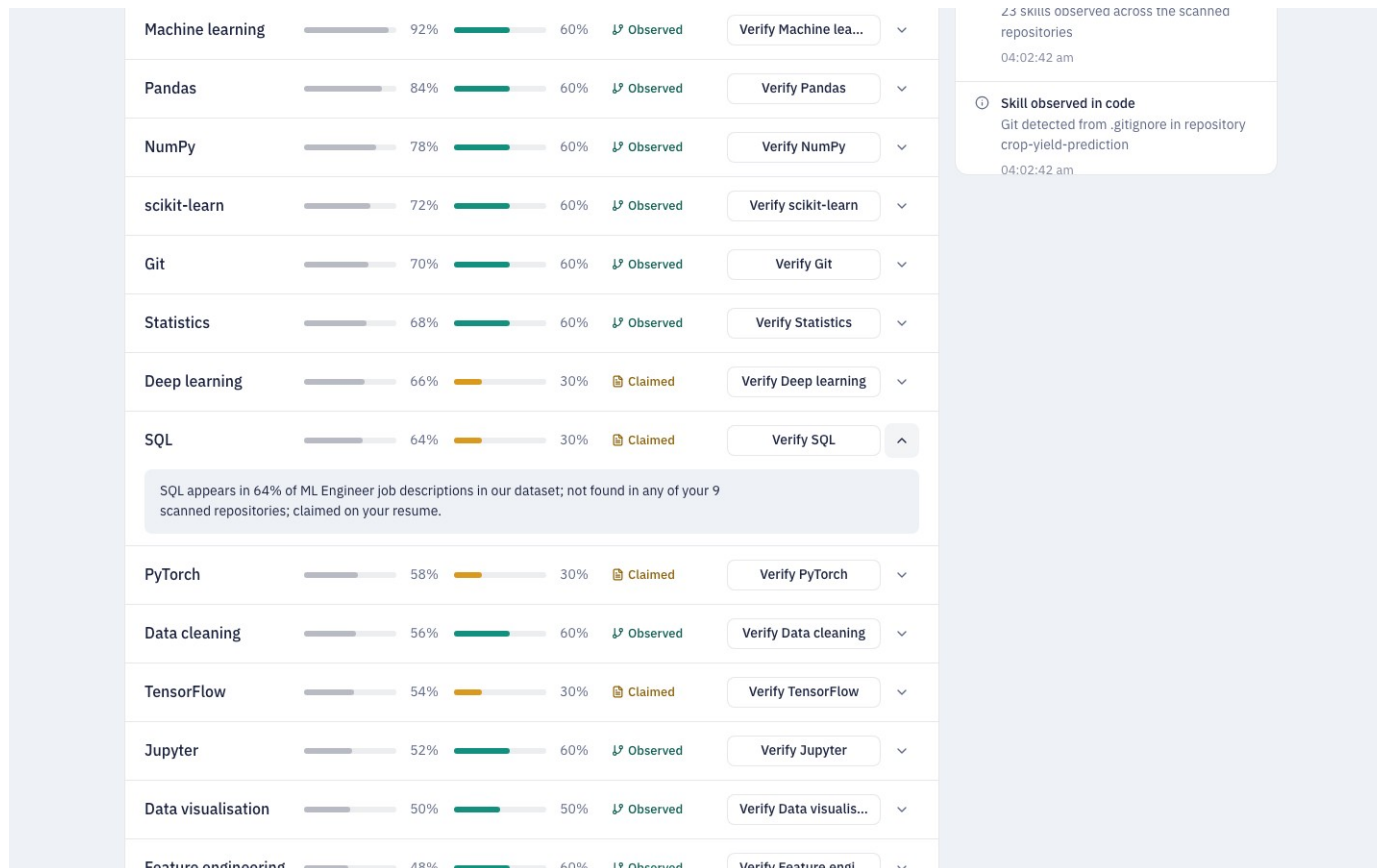
Fill is how much of it is proven

Ceiling: 89

what proving her current evidence would reach, so a gap reads as not shown yet or not learned yet

Every claim is traceable

SQL appears in 64% of ML Engineer listings, sits on her resume, and appears in none of her nine repositories.



Claimed

on the resume only

Observed

found in code, with the repository and file named

Verified

passed an adaptive quiz



Claiming is not proving

SQL was on the resume. Under four adaptive questions it scores 25%, and the badge stays amber.

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DashboardRoadmapLog progress⚙️

Verify SQL

4 questions. Each one adapts: answer correctly and the next is harder, answer wrong and the next is easier. Your score replaces the assumed proficiency for this skill.

Quiz result

21%

 Claimed on SQL

That is below the bar for a verified badge, so SQL keeps its earlier evidence level and its proficiency is now the quiz score. That is useful: it says the gap is real.

Readiness is now **49.1** out of 100 (-0.4)

The change is recorded in your score history.

Open roadmapBack to dashboard

Harder after a correct answer, easier after a wrong one

Scored by difficulty weight, not raw count

The answer never reaches the browser before you answer



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A roadmap that answers why

Eight weeks, eight hours a week, nothing over budget, prerequisites respected.

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Dashboard Roadmap Log progress ⚙

Your roadmap

Ordered by what costs you the most readiness, with prerequisites first. Every item says why it is there and ends in something you can show.

🕒 **Demo profile.** Riya Sharma is a seeded student. Her GitHub scan is replayed from a cached copy in this repository and language model replies come from the recorded cache or the deterministic planner, so nothing here is a live API call.

Review and approve your roadmap. Nothing is fixed until you approve it. Reorder, edit the hours or skip anything that does not fit. [Approve roadmap](#)

Version 1, draft
8 items, 8 weeks, 59 hours in total, 0 done. No week goes past your 8 hour budget. [Rebuild from current gaps](#)

Week 1
5 hours

⚙

Review core ML concepts
Machine learning, 5 hours, in 92% of job descriptions, To do

⬆

Why this is here

Machine learning appears in 92% of ML Engineer job descriptions and is already represented in your repository topics for crop-yield-prediction and 3 more repositories.

Resources

[Google machine learning crash course](#)  course

[Kaggle Learn: intro to machine learning](#)  course

Proof project: Summarise a machine learning workflow

Document the steps from data ingestion to evaluation for one of your existing crop yield prediction projects.

Covers Machine learning

- A written summary of the workflow is produced
- The summary identifies where overfitting could occur

Why is written in numbers, not motivation

Resources come from a whitelist, so no invented links

Every item ends in a proof project with checkable criteria



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It replans when you prove something

Marking a task done changes nothing. Linking the repository moves the score by 8.5 points.

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Dashboard Roadmap Log progress ⚙

Log progress

Readiness is 49.1 out of 100 right now. Anything you record here recalculates it and replans the weeks you have not finished.

🕒 **Demo profile.** Riya Sharma is a seeded student. Her GitHub scan is replayed from a cached copy in this repository and language model replies come from the recorded cache or the deterministic planner, so nothing here is a live API call.

🔗 **What changed**

Readiness moved from 49.1 to 57.6 out of 100, a change of +8.5.

- Model deployment moved from week 4 to week 5, Docker moved from week 3 to week 6.
- Git was added to the plan.
- Statistics was added to the plan.
- NumPy was added to the plan.
- Pandas was added to the plan.
- scikit-learn was added to the plan.
- TensorFlow was added to the plan.
- Deep learning left the plan because it is no longer one of your largest gaps.
- PyTorch left the plan because it is no longer one of your largest gaps.

Link a repository

The fastest way to move your score. We scan the repository, record which file proves which skill, and replan what is left.

Repository URL or owner/name

Repository scanned

On the demo profile this replays the cached scan of ml-deploy-service instead of calling GitHub.

Docker, FastAPI and AWS become observed evidence

The plan rebuilds around what is left

Anything the student edited stays locked



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How it is built

Deterministic where it matters, AI where it helps.

Deterministic core

scoring, gap ranking and skill detection are plain TypeScript, so the same profile always gives the same score

AI at the edges

Sarvam AI writes the explanations and quiz questions, with Groq and Gemini as fallbacks

Everything validated

zod at every boundary, one repair retry, then the next provider, then built in logic

Runs with zero keys

the whole product works on deterministic logic when no provider is available



Stack and engineering

Layer	Choice
Frontend	Next.js 14 App Router, TypeScript strict, Tailwind with a custom token system, Recharts
Agent	Resume and GitHub ingestion, 84 skill taxonomy, deterministic scoring and planner
Models	Sarvam AI sarvam-105b-conversations, then Groq gpt-oss-120b, then Gemini 3.8 Flash
Data	Supabase Postgres in Mumbai, local JSON store when unconfigured
Quality	69 unit tests, typecheck, lint and build in GitHub Actions



Impact and what is next

For a student

one number to argue with, a ranked list of what to fix, a portfolio project per gap

For a placement cell

the same engine across a cohort shows where training should actually go

Next

live job description ingestion, Hindi interface, mentor review as a fourth evidence level

Live demo <https://skillproof-dev-x9.vercel.app>

Source

<https://github.com/shubhamverma-devx/skillproof>